SafeShelf Global

The intelligence layer for every physical product

Identify anything. Discover its twins. Verify its claims. Understand its risks.

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1. Executive summary

SafeShelf Global is an AI platform for physical products. It finds, identifies, compares, verifies and keeps watching products across shopping platforms, manufacturers, sellers and markets.

It is built from five foundational systems:

1. **Universal Product Identity Engine.** Recognises the same physical product across different websites, brands, packaging and listings.
2. **Product Safety Intelligence Engine.** Detects possible hazards from reviews, customer photos, recalls, certifications and incident reports.
3. **Living Product Knowledge Graph.** A database that links products, variants, sellers, manufacturers, components, incidents and safety records, and keeps updating.
4. **Developer API and AI models.** Offers product identification, duplicate discovery, authenticity checks and safety intelligence to other apps and AI agents.
5. **Browser extension and consumer platform.** Gives shoppers useful information while they browse online stores.

Long-term goal: become the shared product identity and intelligence infrastructure for global commerce.

Today every e-commerce platform keeps its own separate product records. SafeShelf instead builds one evidence-backed understanding of each product:

- what it actually is,
- how it relates to other products,
- what is known about its quality and safety.

Two questions about every product

1. **Is it safe?** Hazards, recalls and certifications.
2. **Is it real?** Is it the genuine product, sold by a real shop, at a believable price?

Both answers come from the same identity engine. SafeShelf already knows what the genuine product looks like: its photos, model, normal price range and certificates. So it can tell when a shop copies those photos, prices the product impossibly low, or has only just appeared.

For every shopper. Both answers work with screen readers and the keyboard. They can be read aloud ("Read it to me"), in plain language and in Hindi, so blind, low-vision, elderly and non-English-reading shoppers get the same protection (section 12.1).

What "Is it real?" does not cover: payment, bank, UPI, phishing-SMS or phone-call scams. Those are a different problem, and many other teams already work on them.

The central question

> **"Is this really the same product, can its claims be trusted, and what should I know before buying or selling it?"**

SafeShelf answers with evidence the user can trace, relationships that have been verified, uncertainty stated openly, and a product history that keeps updating.

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2. The problem

2.1 Dangerous products stay on sale

- An **OECD online product-safety sweep found 87%** of banned or recalled products still on sale online (OECD).
- **Australia:** signatories to the ACCC Product Safety Pledge removed about **31,000 unsafe listings** in 2024–25, and the ACCC took action against Amazon in May 2026 (Bird & Bird).
- **India:** BIS raided Amazon and Flipkart warehouses and seized **3,500+ uncertified or fake-ISI items** (Outlook Business). BIS also cancelled registrations over power-bank cells **relabelled as 8,000 mAh** (Communications Today).
- **Energy labels:** from March 2025, BEE star labels must appear on e-commerce listings, and BEE's 2026 regulations require online market surveillance (Enviliance).

2.2 Warnings arrive late

- Buyers often post complaints and **photos of the damage** (melted ports, burn marks, swollen batteries, loose magnets) weeks or months before a product is recalled. By then sales have continued and people may already be hurt.
- **Research:** reading review text alone can predict recalls with about **84% F1** (arXiv 1703.00518; BU SPH).
- **Regulator interest:** the US CPSC lists review-based early warning as an AI use case it wants (CPSC AI use cases).
- **The gap:** no deployed system uses the photo evidence, links rebranded copies of the same product, and hands a reviewer an evidence pack before the recall.

2.3 The same product hides under many names

- One white-label charger, power bank or toy can be sold under dozens of brand names, with new titles, photos and sellers.
- When one listing is recalled or reported, the identical product under another name keeps selling.
- When a listing is taken down, it often comes back as a "new" product.

2.4 Claims can't be checked

- Listings claim certifications (ISI/BIS R-numbers, BEE stars, Australian energy ratings, CE marks) and specs (wattage, mAh, materials) that buyers cannot check.
- **India's official check (BIS Care app):** one number at a time, by hand (BIS).
- **Common tricks:**
  - a genuine number that belongs to a different brand or category,
  - a 5-star claim for a model registered at 3 stars,
  - capacity claims that are physically impossible.

2.5 Fake shops and fake products look real

- Ads on social media lead to brand-new websites selling well-known products at huge discounts.
- These sites copy the genuine product's photos and description. What arrives is a fake, or nothing at all.
- A shopper cannot see that:

  - the photos were copied from the brand's official listing,
  - the price is far below anything the genuine product sells for,
  - the website was set up only a few weeks ago.
- Counterfeits on marketplaces work the same way. The listing borrows the genuine identity, and the product itself is different. India's BIS seizures of fake-ISI goods (section 2.1) are one example.

2.6 Records are fragmented

- Each marketplace, retailer, regulator and manufacturer keeps separate, incompatible product records.
- No one can see that a product's identity, complaints, certifications and recall status belong together.

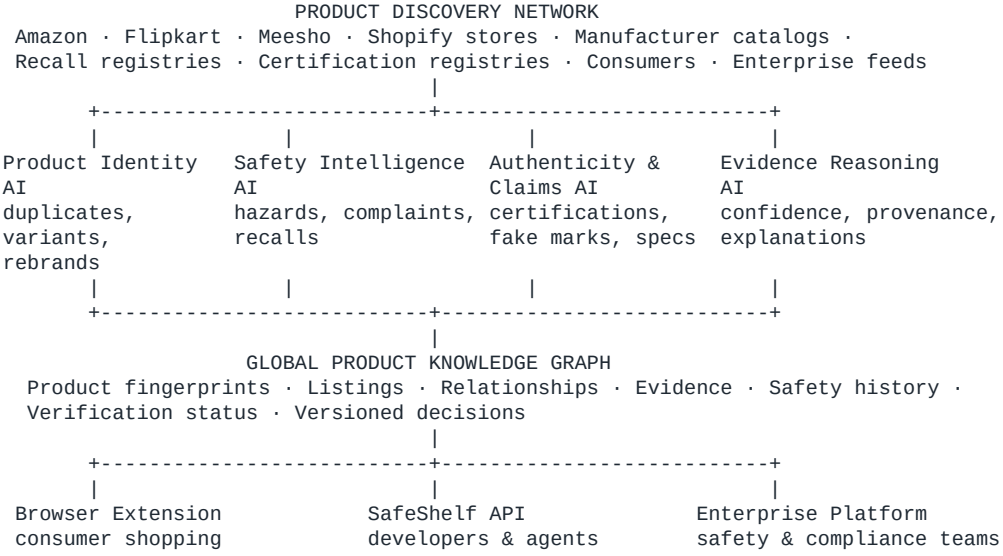
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3. Who SafeShelf serves

User	Persona	Pain today	What SafeShelf gives them
Marketplace product-safety analyst (first paying user)	Anjali, trust-and-safety analyst at an Indian marketplace	Reads complaints by hand, finds rebranded relistings by accident, writes evidence reports from scratch	Ranked hazard queue, linked product families, ready evidence packs
Shopper	Rahul, buying a 65 W charger	Can't tell whether it's genuine, certified or dangerous	Extension shows twins, verified certifications, hazard evidence and recalls
Retailer or manufacturer	Category manager at a D2C electronics brand	Duplicate catalog records, compliance notices, counterfeits using the brand	Catalog deduplication, checks on its own claims, counterfeit monitoring
Regulator or researcher	Product-safety officer at the ACCC or BIS	Sweeps are manual and point-in-time; evidence is scattered	Investigation workspace, traceable evidence exports, aggregate analytics
Developers and AI shopping agents	A shopping app or a Gemini-powered buying agent	No trustworthy, structured product-identity or safety API	`identify`, `duplicates` and `safety` endpoints with evidence IDs

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4. High-level product architecture



All three channels draw on one shared identity and evidence graph. Each has its own access rights and workflow.

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5. The platform's core capabilities (13 modules)

#	Module	Main function	Example
1	Product Discovery	Collect authorised product information	Find charger listings across stores
2	Duplicate Finder	Identify identical or rebranded products	Match two differently branded adapters
3	Product DNA	Build lasting fingerprints from images, text and specs	Match on model, shape, specs and identifiers
4	Safety Radar	Detect emerging product hazards	Spot reports of melted charging ports
5	Recall Intelligence	Match official recalls to affected listings	Find possibly recalled models still advertised
6	Certification Truth	Validate certification claims	Check whether an ISI/BIS licence covers the model
7	Counterfeit & Fake-Shop Watch ("Is it real?")	Flag fakes, impersonation and suspicious shops	Copied photos + impossible price + 19-day-old website
8	Price & Spec Truth	Compare credible specs and prices	Expose conflicting wattage claims
9	Supply-Chain Graph	Identify related manufacturers and components	Investigate chargers that may share a circuit board
10	Safety Passport	Keep the history and verified findings	Track alerts, corrections and investigations
11	Public Evidence Network	Accept structured consumer reports	Upload damage photos and proof of purchase
12	Investigation Copilot	Recommend next checks and actions	Build an evidence package for human review
13	Developer API	Offer all intelligence to other software	Power other shopping apps and AI agents

The modules are not separate apps; they share one product identity and evidence graph.

Module details: how each one works and what Gemini does

1. **Product Discovery**
   - Takes in data only from authorised sources: partner feeds, open datasets, public recall and certification registries, the page a shopper chooses to scan, and uploaded catalogs.
   - Gemini pulls structured attributes (brand, model, wattage, ports, plug type, certifications shown in the images) out of messy titles, descriptions and photos.
2. **Duplicate Finder**
   - Finds candidates by exact identifier, then text and image vector search, then category filters.
   - For unclear pairs, Gemini compares the two products and lists what agrees and what conflicts (same moulding, same label layout, different stated wattage). It then assigns one of five relationship classes (section 7.1).
3. **Product DNA**
   - Gives each product a fingerprint that combines identifiers, normalised attributes, text and image embeddings, and the shape, port layout and label positions in its photos (section 7.3).
4. **Safety Radar**
   - Gemini reads review text in many languages and **looks at review photos**. It classifies hazards using OECD-style hazard types (fire/burn, electric shock, choking, chemical, injury from a mechanical part), estimates severity, and records when each observation was made.
   - Statistics code detects rising complaint rates.

5. **Recall Intelligence**
    - Recall notices (CPSC, ACCC, EU Safety Gate, OECD GlobalRecalls, Health Canada, plus Indian and Japanese notices) are read into structured scope: models, batches, dates and photos.
    - The identity engine matches each notice to listings. It returns **exact model** matches, flags **possible rebrands** for review, and never auto-flags products that are only visually similar.
6. **Certification Truth**
    - Gemini reads the marks and numbers shown in listing images and text: BIS R-numbers and ISI licence numbers, BEE star labels, Australian energy-rating numbers, battery capacity.
    - Code then checks them against the official registries and checks the fields agree with each other: brand, category, model and rating (section 9).
7. **Counterfeit & Fake-Shop Watch ("Is it real?")**
    - **Three small checks**, all built on the identity engine:
        1. **Copied photos.** The listing's images are matched against the genuine product's fingerprint (image embeddings + perceptual hash). If they match the official listing but the seller is not the brand or a known authorised seller, the photos are borrowed.
        2. **Impossible price.** The price is compared with the price range of the genuine product family. Code (not the LLM) flags a price far below the lowest credible price. The threshold is set per category, e.g. under 40% of the lowest authorised price.
        3. **Brand-new website.** For shops outside the big marketplaces, the domain registration date comes from a public RDAP/WHOIS lookup. Code flags a domain registered in the last few weeks.
    - **Counterfeit details:** Gemini also points out conflicts against the genuine product, such as:
        - logo shape,
        - packaging text,
        - a font or language that doesn't match the claimed brand,
        - a manufacturer address that doesn't match the registered one.
    - **How the result is worded:** "Be careful", with the reasons listed. It never says "this is a scam". The checks only describe the listing; the shopper decides.
8. **Price & Spec Truth**
    - Compares the specs a listing claims with the verified fingerprint and the product's family (e.g. "65 W" when its twins are registered at 20 W).
    - Flags capacities that are physically impossible, and prices that suggest a counterfeit.
9. **Supply-Chain Graph**
    - Links products to manufacturers, factories and shared components using registry data, teardown documents and supplier records.
    - When one family has a defect, it shows other products that may share the same component, always marked as a hypothesis.
10. **Safety Passport**
    - A versioned page for each product: identity, twins, certifications, alerts, investigations, corrections and recalls.
    - Old records are never deleted. A correction is added on top.
11. **Public Evidence Network**
    - Consumers submit structured reports: photos, purchase proof, what happened.

- Gemini checks the report is complete and matches the product. It also removes or masks personal information.
- Repeated copies of one report count once.

12. **Investigation Copilot**
    - A Gemini agent with retrieval and tools. It summarises evidence, lists counterevidence, suggests the next checks (ask the seller for test reports, buy a sample, check the registry), and drafts regulator-ready packs: an ACCC mandatory injury report, an EU GPSR notice, or a BIS complaint.
    - A human approves every pack.
13. **Developer API**
    - Versioned REST endpoints (section 11) with authentication, rate limits, usage metering, async jobs and webhooks.

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6. What makes SafeShelf different

Existing tools each do one part:

- comparison tools find cheaper listings,
- recall databases list official recalls,
- review analysers summarise complaints.

SafeShelf joins all three and adds:

- identity resolution (finding the same product under different names),
- a source and history for every piece of evidence,
- product-family relationships that can be explained,
- investigation workflows.

The core differentiator is a product memory that works across markets. A product's verified identity and safety history stay findable even when its title, seller, packaging or marketplace changes.

Competitors and the gap

Who	What they do	GenAI?	Where SafeShelf differs
DeepRecall	Matches listings against 120k+ recalls (CPSC, EU Safety Gate, OECD, ACCC), blocks or removes them, regulator-ready files	Yes	Starts only **after** a recall is announced. SafeShelf aims to warn **before** it, from photo evidence, and resolves rebrands
Recall Shadow (hackathon project)	Turns a recall into a product identity and checks live listings (Gemini + Bright Data)	Yes	Also after the recall; no hazard detection, certifications or graph
Hong Kong EMSD trial (EMSD)	Watches e-commerce listings with OCR and NLP	Partly	Text only, one market, run by the regulator
Academic recall prediction (arXiv)	Review text -> recall risk	No (classic NLP)	No photos, no identity resolution, no workflow
BIS Care app	Check one certification number by hand	No	No automation and no cross-field consistency checks
Catalog matching (Google Shopping, DataWeave, marketplace tools)	Matches duplicate listings for price comparison	Partly	Built for pricing, not safety. No evidence, history or hazard transfer rules
Brand protection (Red Points and others)	Finds counterfeits for brand owners	Yes	Works for one brand, not for the safety of every product

Who	What they do	GenAI?	Where SafeShelf differs
Website reputation checkers (ScamAdviser, Google Safe Browsing, browser security tools)	Score whether a **website** looks risky	Partly	They know about websites, not products. They can't say "these photos belong to Brand X's 65W charger and this price is impossible for that model"
Fake-shop lists (e.g. Austria's Watchlist Internet)	Lists of known fake shops, plus a detector	Partly	Not linked to product identity or safety. Their public lists may help as test data (to be checked)

On "Is it real?": website checkers judge the shop; SafeShelf judges the shop *against the genuine product*. Those are different checks.

Verdict: the parts that come *after* a recall, and duplicate matching on its own, are taken. **Early warning from photos before a recall, with resolution of rebranded twins, certification checks and a human-reviewed evidence pack**, is open. That is the headline. Identity is the engine underneath.

This is a proposed technical and business difference. We do not claim that no competitor offers overlapping features.

Position in the AI Builder Cup

- None of the ~125 public entries (scanned 5 times, Oct 9–11) covers product safety, recalls or certification checks.
- The nearest Retail entries are different problems: payment-slip scam checks, dark-pattern shopper agents, COD order risk.

Part II: Low-level technical design

7. Product identity and duplicate detection

This is the central engineering investment. SafeShelf must tell apart products that are truly identical from products that only look alike.

7.1 Five relationship classes

Class	Meaning	Evidence needed
Exact same model	Identical verified product model	Matching authoritative identifiers (GTIN, registry model number, manufacturer record)
Rebranded equivalent	Same hardware sold under another brand	Manufacturer records, same registry entry, hardware or label comparison
Model variant	Same product family with real differences	Revision, colour, capacity, power rating
Visually similar	Looks alike, but identity is unproven	Embeddings, geometric features only
Shared component	Different products containing a common part	Bill of materials, teardown, supplier documents

Safety rule: a visually similar product must never automatically inherit another product's confirmed recall status. Only *exact same model* inherits recall status. *Rebranded equivalent* can inherit it only after a human verifies the match. All other classes get "related evidence", never "recalled".

7.2 Matching pipeline

Step	Name	What happens
01	Acquire listing	Collect the authorised URL, title, images, identifiers and specs
02	Normalise	Standardise brand names, units (W, mAh, V), model strings and image metadata
03	Generate fingerprint	Build text, image and attribute representations
04	Retrieve candidates	Search exact IDs, vector indexes and category blocks
05	Verify candidates	Gemini or a specialised matcher checks what agrees and what conflicts
06	Resolve identity	Match, link as a possible relative, or send to human review
07	Update graph	Store the relationship as a new version, with evidence and confidence

7.3 Global product fingerprint

A fingerprint is a structured identity record, not just one image embedding.

```
{
  "product_id": "SS-P-10291",
  "category": "usb_charger",
  "brand": "ExampleBrand",
  "model": "CX-65",
  "gtin": null,
  "attributes": {
    "power_watts": 65,
    "ports": 2,
    "plug_type": "Type C"
  },
  "visual_features": {
    "port_layout": "2x USB-C stacked",
    "label_text_hash": "a91f...",
    "housing_shape": "rounded_rectangle"
  },
  "certifications_claimed": ["BIS R-41XXXXXX"],
  "image_embedding_ref": "EMB-921",
  "text_embedding_ref": "EMB-922",
  "identity_version": 3,
  "verification_status": "provisional"
}
```

Embeddings only *propose* candidate matches. Actual identity decisions rest on authoritative identifiers, consistent attributes and evidence.

7.4 Gemini verification prompt (structured output)

For each candidate pair, Gemini receives both listings (images and text) and returns:

- `agreements[]`: e.g. same port layout, same label text, same registry number.
- `conflicts[]`: e.g. different stated wattage, different plug.
- `proposed_class`: one of the five classes.
- `evidence_spans`: which image or text supports each point.
- `needs_human`: `true` whenever identifiers are missing or there are conflicts.

Code then applies the final decision rules, so the LLM never decides a recall transfer on its own.

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8. Product safety detection and reasoning

The safety pipeline has two separate goals:

1. find reported hazards;
2. decide whether the evidence justifies an investigation.



8.1 What each stage does

- **Text hazard extraction (Gemini, many languages).** Reads a review and decides: hazard present? Which type and mechanism ("port melted while charging overnight")? Was anyone injured? Is it the user's own experience, hearsay or a joke?
- **Image damage analysis (Gemini vision).** Decides whether the photo shows real damage, what part is damaged, and whether the damage looks like heat, impact or wear. It also checks that the photo shows *this* product and not some other item.
- **Duplicate and spam filtering.** Copies of the same complaint, reposted photos (found by image hashing) and review-farm patterns count as **one** piece of evidence.
- **Certification and recall verification.** Is the product already recalled? Is the certification it claims real and does it apply to this model?
- **Aggregation and prioritisation.** Each product family gets a score from:
  - severity,
  - number of independent reports,
  - change in complaint rate (statistical change detection),
  - photo evidence,

- certification problems,
- how vulnerable the users are (e.g. children's products).

8.2 What every alert keeps

- Hazard type and a severity hypothesis.
- References to the original evidence, with the date each was observed.
- Why the evidence supports the flag.
- Counterevidence and anything still uncertain.
- The product's identity and how confident we are in its relationships.
- Whether it has been checked against a registry or recall.
- The human reviewer and the decision history.

AI findings are always shown separately from verified regulatory or laboratory findings, with different labels and colours.

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9. Certification Truth: details

Claim	Registry	What we check
ISI mark / BIS R-number (India)	BIS CRS registry / BIS Care	Number exists, is active, belongs to **this brand and category**, covers this model
BEE star label (India)	BEE star-label registry	Registered stars for this model match the claimed stars
Australian energy rating	Energy Rating registry, open CSV/API	Model, brand and star rating all agree
Battery capacity (mAh/Wh)	Registered cell data, physics checks	Claim is plausible for the size and weight; matches the registered cells
CE / RCM marks	Declarations of conformity, where public	Mark is present and consistent with the stated manufacturer

Every check returns `verified`, `mismatch` (with the field that differs), or `unknown` (registry can't be reached or the number can't be read). The system never guesses.

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10. Database architecture

Starting stack:

- **PostgreSQL with** `pgvector` on Cloud SQL, for identity, evidence and vector search.
- **Cloud Storage** for original photos and documents.
- **BigQuery** for trends and analytics.

A dedicated graph database isn't needed on day one; relationship tables and recursive queries are enough.

10.1 Core entities

Table	Important fields
`products`	id, brand, model, category, identity_status, created_at
`product_families`	id, family_name, manufacturer_id
`manufacturers`	id, name, country, registry_ids
`sellers`	id, marketplace, seller_ref, domain, domain_registered_at, is_authorised, first_seen
`price_observations`	product_id, listing_id, price, currency, observed_at

Table	Important fields
`trust_checks`	listing_id, copied_photo_match, matched_product_id, price_ratio_to_floor, domain_age_days, result, checked_at
`listings`	id, product_id, seller_id, marketplace, url, title, captured_at
`product_fingerprints`	product_id, text_vector, image_vector, attributes_json, version
`product_relationships`	source_id, target_id, relationship_type, confidence, status, evidence_ids, version
`product_attributes`	product_id, attribute, value, unit, source, verified
`hazard_signals`	id, product_id, hazard_type, severity, status, first_seen, last_seen
`evidence`	id, source_type, source_ref, storage_uri, observed_at, captured_at, hash, pii_status
`certifications`	product_id, authority, number, claimed_value, registry_value, validation_status, checked_at
`recalls`	id, authority, notice_date, scope_json, hazard, url
`investigations`	id, product_family_id, status, assigned_reviewer, decision, decided_at
`reports`	id, reporter_ref (hashed), product_id, description, evidence_ids, status
`audit_events`	entity_id, entity_type, change_type, actor, before, after, timestamp

Join tables for many-to-many links:

- `evidence_claims`: one piece of evidence can support several claims.
- `recall_products`: one recall can cover many models or batches.
- `investigation_evidence`
- `product_components`

10.2 Engineering rules

- **Immutable evidence.** Original evidence and its source metadata are kept, with access controls and retention policies.
- **Versioned conclusions.** If a flag is disproved, the history stays and a correction is added.
- **No unsafe merging of identities.** Different products stay provisionally linked until there is enough verification.
- **Source rights.** Product content, review photos and feeds are accessed and kept only when authorised.
- **Privacy.** Personal information in customer reports is removed or restricted.
- **Multi-tenant access.** One enterprise customer never sees another's confidential evidence by default.

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11. API design

The intelligence is offered through reusable, versioned APIs, not built only into the extension.

Endpoint	Function
`POST /v1/products/identify`	Resolve a listing, image or identifiers to a product
`POST /v1/products/duplicates`	Find verified and possible duplicates
`GET /v1/products/{id}`	Get the product's identity
`GET /v1/products/{id}/safety`	Get evidence-backed safety findings
`GET /v1/products/{id}/relationships`	Explore variants and related products
`GET /v1/products/{id}/history`	See product changes and safety events
`GET /v1/products/{id}/certifications`	Get verification results

Endpoint	Function
`POST /v1/listings/trust-check`	"Is it real?": copied photos, impossible price, website age for any listing URL or page data
`POST /v1/reports`	Submit a consumer or analyst report
`GET /v1/investigations/{id}`	See an investigation's status (if authorised)
`POST /v1/analyses`	Start an analysis that may run for a long time

11.1 Example duplicate-detection response

```
{
  "query_product_id": "SS-P-10291",
  "matches": [
    {
      "product_id": "SS-P-20019",
      "relationship": "exact_model",
      "status": "verified",
      "evidence_ids": ["EV-101", "EV-102"]
    },
    {
      "product_id": "SS-P-30118",
      "relationship": "possible_rebrand",
      "status": "unverified",
      "confidence": 0.82,
      "evidence_ids": ["EV-108"]
    }
  ],
  "graph_version": 12
}
```

11.2 Example safety response

```
{
  "product_id": "SS-P-10291",
  "official_status": { "recalled": false, "checked": ["CPSC", "ACCC", "EU Safety Gate"] },
  "ai_findings": [
    {
      "hazard_type": "fire_burn",
      "mechanism": "USB-C port overheating during charging",
      "independent_reports": 9,
      "with_photos": 4,
      "first_seen": "2026-07-02",
      "trend": "rising",
      "status": "under_review",
      "evidence_ids": ["EV-201", "EV-207", "EV-215"],
      "counterevidence": "Most reports involve third-party cables"
    }
  ],
  "certifications": [
    { "authority": "BIS", "number": "R-41XXXXXX", "result": "mismatch", "field": "brand" }
  ],
  "label": "AI finding - not a regulatory determination"
}
```

11.3 Example "Is it real?" response

```
{
  "listing_url": "https://example-deals-store.test/65w-charger",
  "matched_product_id": "SS-P-10291",
  "checks": {
    "copied_photos": { "result": "flag", "matches_official_listing": true, "seller_authorised": false },
    "price": { "result": "flag", "price": 299, "lowest_credible_price": 1399, "ratio": 0.21 },
    "website_age": { "result": "flag", "domain_registered_days_ago": 19 }
  },
  "summary": "Be careful: photos copied from the official listing, price far below the genuine product, website only
  "safety_of_genuine_model": "9 hazard reports under review, no recall",
  "label": "Evidence for the shopper - not a legal finding"
}
```

`confidence` is shown here only as an illustration. In production it must be a measured, calibrated model output, not an arbitrary similarity score.

11.4 Platform requirements

- API keys or OAuth
- Rate limits and quotas
- Pagination
- Async jobs and webhooks
- Usage metering
- Request validation
- Audit logging

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12. Browser extension design

User experience (all data below is fictional demo data):

> **65W USB-C Charger** · Identity SS-P-10291 · (!) Unverified hazard report
> **12** potential related listings · **3** hazard evidence items
> Actions:
> - Find duplicates and variants
> - View safety findings and sources
> - Compare product specifications
> - View product safety timeline
> - Report a suspected problem

"Is it real?" panel (also fictional demo data), shown on any shop page, including unknown websites:

> **Is it real?** Be careful.
> - Photos copied from the official Brand X listing; this shop is not an authorised seller
> - Price Rs 299; the genuine product sells for at least Rs 1,399
> - Website created 19 days ago
> **Is it safe?** The genuine model has 9 overheating reports with photos (under review). No recall.

Design rules:

- Ask for **minimal permissions**. Run only when the shopper opens a supported product page or asks for a scan.
- **Never collect unrelated browsing activity silently.**
- Send only the product page's details to the API, never the shopper's identity.
- Show verified facts (green), AI findings (amber, "under review") and official recalls (red) in different styles.
- **First version:** Chrome plus two shopping sites. Add Firefox, Edge and more stores after the matching pipeline has proven itself.
- **Accessibility:** built in from the start, see section 12.1.

12.1 Accessibility: SafeShelf for every shopper

Why it matters here. The evidence SafeShelf relies on is mostly visual:

- buyer photos of melted or burnt products,
- safety marks and numbers printed on the pack,
- price and seller details shown in images.

Blind and low-vision shoppers using screen readers cannot see any of this, so they are more exposed to fake and dangerous products than anyone else. SafeShelf already turns these images into words, so it can also

speak them.

What we build:

Feature	Who it helps	How
Accessibility basics	Blind, low-vision and colour-blind shoppers, keyboard-only users	Every panel works with the keyboard alone. All buttons and results have screen-reader labels (ARIA). Contrast meets WCAG 2.2 AA, and text can be enlarged. Warnings always use **an icon and words as well as colour** (green / amber / red is never the only signal). Focus order follows the reading order
"Read it to me"	Blind, low-vision and elderly shoppers, people who find reading hard	One button (or a keyboard shortcut) reads a short spoken summary using Gemini TTS, for example: *"This charger looks genuine. Nine buyers reported overheating, under review. Not recalled."*
Plain-language mode	People with cognitive disabilities, first-time online shoppers	Gemini writes the summary in short sentences with no jargon. Each warning has a "Why?" button with one simple reason and the evidence behind it
Indian languages	Shoppers who don't read English	The summary and voice in Hindi first, with more Indian languages later
Ask by voice (stretch goal)	Blind and elderly shoppers	Gemini Live: the shopper asks "Is this charger safe?" or "Is this shop real?" and hears the answer, grounded in the same evidence and API results

Rules:

- **The spoken answer never says more than the evidence.** It uses the same wording rules as the screen: "be careful", "under review", never "scam" or "dangerous" unless there is an official recall.
- **Short first, detail on request.** The first spoken sentence gives the verdict. Detail is read only when the shopper asks.
- **No extra permissions.** Voice runs only when the shopper presses the button or the shortcut.

How we check it:

- **Google Lighthouse** accessibility score of 95 or higher on every screen.
- An **axe** automated scan with no serious issues.
- **Manual tests** with the NVDA screen reader (Windows) and TalkBack (Android).
- Keyboard-only walkthroughs of the four hero screens.
- On the deck: "Meets WCAG 2.2 AA (self-tested)".

Scope: accessibility is a layer on top of SafeShelf, not the main idea. SafeShelf stays a Retail product that everyone can use.

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13. AI and machine-learning architecture

We won't train a large language model from scratch. Instead:

- foundation models (Gemini) do the understanding;
- smaller specialised models help;
- deterministic code does the verification.

AI component	Approach	Output
Product recognition	Gemini multimodal + Vertex AI multimodal embeddings	Product attributes and identity candidates

AI component	Approach	Output
Duplicate matching	Vector retrieval + trained reranker (Gemini on unclear pairs)	Related products and evidence
Hazard extraction	Gemini + specialised classifier	Structured incident signals
Image damage analysis	Gemini vision with structured output	Damage type, part, consistency with the product
Product graph reasoning	Graph algorithms first, graph ML later	Possible shared defects and product relationships
Certification verification	OCR/vision (Gemini) + official registry lookup	Verified, mismatched or unknown
"Is it real?" checks	Image embeddings + perceptual hash for copied photos; price-floor rules; RDAP domain age; Gemini for counterfeit detail conflicts and the plain-language summary	Flags with reasons, never a verdict
Early-warning detection	Statistical change detection + calibrated models	Investigation priority
Investigation copilot	Gemini with retrieval and tools (ADK agent)	Evidence-grounded explanation and next steps

Models:

- **Gemini 3.x on Vertex AI.** The model ID is kept in an environment variable. Gemini 2.5 retires on 16 Oct 2026.
- **Gemma** for cheap pre-filtering of large review volumes.
- **Vertex AI embeddings** for text and images.

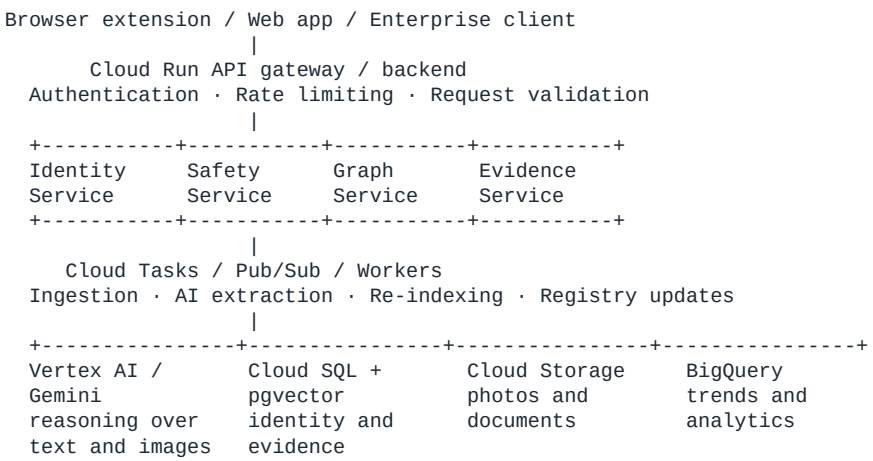
SafeShelf-Tiny (later): a compact product-identity and hazard model, trained only on reviewed examples we are allowed to use for training. It lowers cost at scale and can run close to the data.

Original research directions:

- learning product identity and shared hazard mechanisms together;
- discovering product families nobody knew existed;
- choosing which extra piece of evidence would most improve an uncertain safety conclusion (active evidence acquisition).

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14. Google Cloud infrastructure



From the start: Cloud Monitoring and Logging, Secret Manager, encryption at rest and in transit, backups, IAM access policies and audit logging.

Later, when real workloads justify it: enterprise-scale graph infrastructure and more specialised retrieval (Vertex AI Vector Search).

15. How the database keeps growing

SafeShelf improves through a controlled loop of learning and verification:

1. A new listing or report arrives.
2. The identity engine finds possible matches.
3. Models extract attributes, claims and hazard signals.
4. The system removes duplicates and judges how credible each source is.
5. Verified and provisional relationships are stored separately.
6. Analysts confirm, reject or correct uncertain claims.
7. Indexes and calibrated models improve from the reviewed outcomes.

Rules:

- Repeated copies of the same complaint never count as independent evidence.
- There is a transparent correction process. Sellers, brands and users can challenge wrong product links and safety warnings, and each challenge has a tracked resolution.

16. Large-scale system design

Stage	Scope	Architecture priorities
Prototype	100–500 curated products	Simple database, Gemini, extension, API
Pilot	10,000–100,000 product records	Vector indexing, queues, deduplication, automated testing
Growth	1–10 million listings	Partitioning, batch ingestion, cached results, faster retrieval
Enterprise	Tens of millions of listings	Distributed ingestion, multi-tenancy, SLOs, data governance
Global network	Many countries and categories	Regional compliance, identity federation, multilingual models

These are planning targets, not tested performance claims.

Cheap first, expensive last: match exact identifiers first, then retrieve plausible candidates by vector search, and call Gemini only for unclear cases and high-value investigations. This keeps running costs manageable.

17. Trust, legal and privacy

- **Evidence, not verdicts.** SafeShelf never says a product "is dangerous". It says "N independent reports with photos of overheating, under review", and shows the evidence.
- **Defamation risk.** Public warnings appear only once there is a verified or official finding. Unverified AI findings stay in the analyst and enterprise view. The public demo uses only recalls that are already official.
- **"Is it real?" wording.** Shoppers see "Be careful" and the facts behind it: copied photos, the price gap, website age. SafeShelf never calls a shop or seller a scam. A genuine new shop or a real clearance sale can trigger a flag, so the reasons are always shown.
- **Right of reply.** Sellers and brands can respond, and corrections stay visible.
- **Data rights.** No scraping against a site's terms. Sources:
  - partner feeds,
  - open datasets (e.g. Amazon Reviews 2023),

  - public registries,
  - the page a user chooses to scan,
  - uploaded catalogs.
- **Privacy.** Reporter identities are hashed. Faces, names and addresses are blurred or removed from photos and text before storage.
- **Security.** Tenant isolation, least-privilege IAM, encryption, audit trails.

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18. Business model

- **Consumers: free core safety access.** Official recalls, verified warnings, evidence sources and basic duplicate matches. Advanced comparison could be a paid plan.
- **Developers: usage-based API.** Charges for product resolution, duplicate matching, large-scale enrichment and advanced intelligence queries.
- **Retailers and manufacturers: enterprise SaaS.** Catalog deduplication, compliance checks, product-risk monitoring, counterfeit investigations and workflow automation.
- **Regulators and research organisations: specialised tools.** Authorised investigation workspaces, traceable evidence exports and aggregate safety analytics.

Strongest first paid product: catalog matching and deduplication. Businesses need it every day, which is more often than recall incidents happen, and safety intelligence then becomes the higher-value add-on. The regulatory pressure on marketplaces (ACCC Pledge, BIS raids, BEE online surveillance, EU GPSR) gives a reason to buy now.

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19. How to measure success

Identity and safety need separate accuracy measures, because a wrong duplicate match could carry a hazard warning onto an unrelated product.

Metric	What it measures
Exact-match precision	Share of claimed identical products that really match
Duplicate recall	Share of known duplicates found
Relationship calibration	Whether predicted confidence matches observed correctness
Hazard alert precision	Share of safety alerts that deserve investigation
Early detection lead time	How much earlier signals appear than official notices
Source traceability	Share of claims linked to accessible evidence
API latency	Response time, cached vs new analyses
Cost per analysis	Compute, model inference and data costs
Analyst efficiency	Time saved reviewing one safety case
Correction time	How fast disproved flags are resolved

For the hackathon, the priorities are identity precision, evidence quality and **one compelling historical safety investigation**, rather than wide coverage.

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20. The complete end-to-end experience

A shopper opens a product page for a charger.

1. SafeShelf recognises the listing and extracts the model, brand, specs and images.
2. The identity engine searches the network for exact models, rebrands and visually similar products.

3. The knowledge graph retrieves safety history, product relationships, official recall records and certification information.
4. The reasoning engine explains what it found, keeping verified facts clearly apart from suspicions.
5. The extension shows the results: duplicate listings, spec conflicts and any relevant safety evidence.
    - On an unknown shop it also answers **"Is it real?"**: are the photos copied, is the price impossible, is the website brand new?
6. The user can submit a report or look at the evidence behind a warning.
7. The same data is available through the API, subject to permissions and data licensing.
8. Verified new findings update the network and help future investigations and searches.

Same intelligence, different users:

- **Retailer:** the same intelligence at catalog scale.
- **Regulator:** an investigation view built on evidence.
- **AI shopping agent:** a structured API response.

Everyone uses the same core system, with different access rights and workflows.

Part III: Hackathon build (AI Builder Cup, submit 17 Oct 2026)

The full platform above is the vision. For the hackathon we build **one working slice through every layer**, so judges see the whole system in action, and put the rest on the roadmap slide.

21. What we demo

Hero story: "We would have caught it N weeks earlier."

1. We pick a past, official CPSC recall of a charger or power bank, with reviews in the Amazon Reviews 2023 dataset.
2. **Timeline replay:** we play the reviews in date order. SafeShelf's Safety Radar raises a flag on the day the evidence crosses the threshold, with photos of melted ports, and the official recall date appears weeks later.
3. **Twins:** the Duplicate Finder shows the same product under other brand names. One is an *exact model* (inherits the recall). One is a *possible rebrand* (sent for human review). One is *visually similar* (shows related evidence only, never "recalled").
4. **Certification Truth:** a listing claims a 5-star energy rating, but the Australian registry shows 3 stars, so it's a mismatch, with the registry record shown.
5. **Investigation Copilot:** it drafts the evidence pack and regulator report. The analyst approves it.
6. **Is it real? (about 20 seconds):** we open a mock "fake shop" page we built ourselves, clearly labelled as a demo. It sells the same charger for Rs 299. The extension shows that the photos are copied from the official listing, the price is far below the genuine product, and the website is 19 days old. It then adds that the genuine model also has hazard reports.
7. **For every shopper (about 15 seconds):** a screen reader is running on a shopping page. The shopper presses one key, and SafeShelf says in Hindi: *"Be careful. This shop copied the brand's photos, and the price is too low to be real."*
8. **Extension and API:** the Chrome extension shows the same findings on a demo product page. One `curl` to `/v1/products/{id}/safety` returns the same JSON.

22. Hackathon scope (every module shown, depth chosen)

Module	Hackathon depth
Product Discovery	Open datasets (Amazon Reviews 2023 metadata + review images), CPSC recalls API, Australian energy-rating CSV, a curated set of 300–500 products
Duplicate Finder + Product DNA	**Full:** fingerprints, vector search in pgvector, Gemini pair verification, 5 relationship classes
Safety Radar	**Full:** Gemini hazard extraction from text and photos, dedup, change detection, timeline replay
Recall Intelligence	**Full:** CPSC notices parsed and matched to products
Certification Truth	**Full** for the Australian energy-rating registry; BIS shown as a mock connector
Counterfeit & Fake-Shop Watch ("Is it real?")	**The 3 checks in full** (copied photos, impossible price, website age) + basic counterfeit detail conflicts. Nothing more: no payment, phishing or call scams
Price & Spec Truth	Basic: claimed spec vs fingerprint conflicts
Supply-Chain Graph	Shown as a graph view of the families we built (no component data)
Safety Passport	**Full:** a versioned product page with history and corrections

Module	Hackathon depth
Public Evidence Network	Simple report form with photo upload and PII masking
Investigation Copilot	**Full:** ADK agent drafting the evidence pack, human approval
Developer API	`identify`, `duplicates`, `safety`, `history`, `trust-check` on Cloud Run with API key
Browser extension	Chrome, one demo product page
Accessibility (section 12.1)	**Full:** WCAG 2.2 AA basics, "Read it to me" (Gemini TTS), plain-language mode, Hindi summary. **Stretch:** ask by voice (Gemini Live)

23. Evaluation (performance slide)

- **Early-warning backtest:** 20+ past CPSC recalls with matching Amazon reviews. Report the median lead time between our flag and the recall date, plus precision on a set of products that were never recalled (false alarms).
- **Identity:** ~200 hand-labelled product pairs. Report exact-match precision and duplicate recall per relationship class.
- **Certification:** seeded tampered listings checked against the Australian registry. Report precision and recall (exact ground truth).
- **Is it real?** We build about 50 fake-shop pages ourselves:
  - copied photos,
  - prices cut to 20–40% of the genuine price,
  - a few changed specs.

We add 50 genuine listings, including real discounts and new authorised shops. Then we report precision and recall for each of the 3 checks. The labels are known because we made the pages.

- **Accessibility:**
  - Lighthouse accessibility score of at least 95 on every screen,
  - axe scan with no serious issues,
  - NVDA and TalkBack walkthroughs,
  - time for a screen-reader user to get the verdict on a product page (target: one key press, under 10 seconds).
- **Analyst time:** building an evidence pack by hand vs with SafeShelf.

Make-or-break test (first thing on Oct 12, about 2 hours): do hazard reviews and photos appear at least 2 weeks before the recall date on enough recalls? If yes, build SafeShelf. If no, stay with ClearSight.

24. How this scores against the judging criteria

Criterion	Weight	How SafeShelf earns it
Technical merit & GenAI	40%	Gemini does real work: image damage analysis, multilingual hazard extraction, pair verification, certification reading, an agentic copilot. Code makes the final decisions; there's a full stack on Cloud Run and a measured backtest
Problem & impact	25%	Lives and injuries; OECD 87%; ACCC and BIS enforcement. It serves the marketplace, the shopper and the regulator. "Is it real?" also helps every shopper who clicks a too-good-to-be-true ad
Innovation	25%	Early warning *before* the recall, from photos, plus the rule that only verified twins inherit a recall

Criterion	Weight	How SafeShelf earns it
UX	10%	Timeline replay, the twin view with colour-coded certainty, an extension in the shopper's flow. **Accessible to blind, low-vision and elderly shoppers** (screen reader, "Read it to me", Hindi, plain language)

Rough score: Tech 4.5, Impact 4.5, Innovation 4.5, UX 4.0 gives **4.45**, and about 4.5 if the backtest shows a lead time of at least 2 weeks.

Accessibility adds roughly **+0.03 to +0.05** overall (UX up about 0.25–0.5, a small Impact lift), and makes us a stronger contender for the separate **Best UI/UX** prize.

"Is it real?" adds roughly **+0.05 to +0.1**, mainly to Impact, as long as it stays at 3 small checks. Grown into a full scam app, it would overlap the many scam entries and pull the score down.

25. Build plan

Day	Work
Oct 12	Make-or-break backtest. Confirm Gemini 3.x model IDs. Load datasets into Cloud SQL and Cloud Storage
Oct 13	Fingerprints, embeddings, candidate search, Gemini pair verification, relationship rules
Oct 14	Safety Radar (text and photos), dedup, change detection, recall matching, timeline replay
Oct 15	Certification Truth, **"Is it real?" checks** (copied photos, price floor, website age) and the 100-page test set, Investigation Copilot (ADK), API endpoints
Oct 16	Safety Passport page, Chrome extension (both panels), mock fake-shop page, report form. **Accessibility pass:** ARIA labels, keyboard, contrast, "Read it to me" (Gemini TTS), Hindi summary, Lighthouse/axe/NVDA checks. Evaluation runs, polish, freeze. (Ask by voice only if time is left)
Oct 17	Demo video, deck, submit

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26. Development roadmap

Phase	Goal	What gets built
1. Hackathon MVP	Show the core innovation	Extension, fingerprints, duplicate discovery, safety flagging, evidence view, developer API, on a small curated dataset
2. Private pilot	Prove real-world value	Better matching, more data sources, review workflows. Measure precision, false positives and cost per product
3. Commercial platform	Sell it	Enterprise dashboards, catalog feeds, SLAs, webhooks, API billing, certification checks, safety monitoring
4. Specialised models	Lower cost, higher accuracy	SafeShelf-Tiny trained on validated relationships and hazard evidence; more languages and image types
5. Global infrastructure	Become the network	A larger identity network, trusted data partnerships, supplier-level evidence, many markets

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27. Risks and hard questions

Risk / judge question	Answer
"DeepRecall already does this."	DeepRecall acts after a recall. We warn before it, from photos, and resolve rebranded twins. The timeline replay shows the difference
A wrong match puts a hazard on an innocent product	Only exact models inherit a recall; rebrands need a human; visually similar never inherits. Identity precision is measured separately
False alarms	Every alert is "under review" until a human approves it; we report precision on products that were never recalled
Data rights	Open datasets, public registries, partner feeds and user-initiated scans only

Risk / judge question	Answer
Defamation	The public view shows only verified or official findings; there's right of reply and a correction log
Review photos are sparse in some categories	Text evidence still works; photos increase confidence but aren't required
Some registries (BIS) are behind a captcha	Australia's open registry for the demo; BIS via an official API or partnership later
Too big for 6 days	We build one deep slice through every layer and show the rest on the roadmap
Accessibility done badly (labels missing, voice too long)	Build it in from day one instead of adding it at the end; run Lighthouse and axe on every screen; keep spoken answers to one sentence first
"Isn't this just another scam checker?"	No. It covers only deception about the product (copied photos, impossible price, brand-new shop, fakes), checked against the genuine product's identity. Payment, phishing and call scams are out of scope
A real shop gets flagged (new shop, genuine sale)	The result says "Be careful" with the reasons, never "scam". Authorised-seller lists and the price history reduce false flags, and the test set includes real discounts and new shops

28. Final product definition

SafeShelf Global

- **Category:** AI-powered product identity, safety intelligence and commerce infrastructure.
- **Mission:** make physical products identifiable, comparable and accountable across the global digital marketplace.
- **Core technology:**
  - multimodal product fingerprints,
  - duplicate resolution,
  - an evidence-backed product knowledge graph,
  - specialised safety reasoning,
  - open developer interfaces.
- **Main products:** browser extension, developer API, enterprise intelligence platform, and a product database that keeps growing.
- **Long-term vision:** a trusted product intelligence network that helps:
  - consumers buy with confidence,
  - developers build better shopping experiences,
  - retailers keep accurate catalogs,
  - safety organisations find dangers earlier.

The most important strategic decision: build SafeShelf around **product identity and trusted evidence**, not a single AI safety score.

- Duplicate discovery gives everyday usefulness.
- Safety investigation gives real differentiation.
- The API and the growing knowledge graph are the foundation of a scalable technology company.

The MVP can stay focused. The long-term platform can grow much larger, but only if identity accuracy, data rights and evidence reliability are solved well.